

Marabūt's Theory of Gravity: The Illusion of Spacetime

Redefining Time Dilation as Thermodynamic Fluid Impedance

Christopher Berry Marabūt^{*1} and Angelina Lopez Marabūt^{†1}

¹Independent Researchers

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Abstract

General Relativity revolutionized gravity by modeling it as the curvature of a four-dimensional “spacetime” fabric. However, this geometric abstraction fails to provide a mechanical, deterministic explanation for gravitational phenomena at the quantum level. This paper utilizes the Electron Flow Model (EFM) to replace the geometric abstraction of spacetime with the physical thermodynamics of the universal vacuum flux. We demonstrate that gravity is the macroscopic flow of fluid into Mara Class sinks, and that “time dilation” is not the warping of a temporal dimension, but the physical slowing of atomic oscillators due to increased thermodynamic impedance within a dense fluid medium.

1 Introduction: The Rubber Sheet Illusion

Legacy physics continuously illustrates gravity using the analogy of a heavy bowling ball resting on a rubber sheet, creating a depression that alters the path of smaller marbles. However, this analogy suffers from a fatal, circular flaw: it relies on an external gravitational force pulling the bowling ball “downward” to explain the very concept of gravity itself.

While Albert Einstein's General Relativity provided an unprecedented, highly accurate mathematical framework for predicting macro-gravitational behavior, it replaced physical mechanics with pure geometric abstraction. The concept of an empty, bendable “fabric” of spacetime offers no underlying physical mechanism for how mass grips the vacuum or why time slows down.

The Electron Flow Model (EFM) decisively breaks this century-old geometric illusion. The EFM rejects the existence of an empty, pliable void. Instead, space is a highly pressurized, supersaturated fluid medium [marabut2026m1]. What Einstein modeled mathematically as curved four-dimensional geometry, the EFM translates directly into mechanical fluid density gradients and deterministic hydrodynamic flow [marabut2026m6]. This paper demonstrates that gravity and time dilation are not the warping of space and time, but the physical, thermodynamic consequences of a fluid universe in motion.

^{*}ORCID iD: 0009-0001-9759-9587 — Email: chris.marabut@nestlink.com

[†]ORCID iD: 0009-0007-9937-6145 — Email: angelina.marabut@nestlink.com

2 Gravity as Fluid Flow, Not Curvature

As established in earlier EFM papers, an EVGS (Electron Void Gravitational Sphere) is not a puncture or singularity in spacetime [marabut2026m10]; it is a massive thermodynamic sink [marabut2026m8]. Gravity is simply the macroscopic river of vacuum flux rushing inward to seek equilibrium. Objects fall toward an EVGS not because geometry dictates their path, but because they are physically carried by the localized eddy currents of the pressurized fluid river.

3 Time Dilation as Hydrodynamic Impedance

Legacy physics claims that “time” itself slows down near a massive object. The EFM argues that time is an absolute, universal progression. What slows down are the physical, mechanical interactions of atoms. As an atomic clock moves closer to an EVGS, the density and pressure of the inward-flowing vacuum flux increase drastically [marabut2026m4].

This denser fluid environment creates a higher localized thermodynamic impedance (τ). The energy cost of every atomic transition is raised, and the mechanical oscillation of the clock's constituent atoms literally slows down due to increased fluid drag. Time does not bend; physical mechanisms simply encounter more resistance.

Derived directly from the Unified Master Equation, the fluid density gradient $\nabla\rho_{\text{flux}}$ dictates that thermodynamic impedance (τ) scales with the gravitational potential. The EFM local processing time (Δt_{local}) is governed by the universal progression ($\Delta t_{\text{universal}}$) divided by the fluid drag ratio [marabut2026m2]:

$$\Delta t_{\text{local}} = \frac{\Delta t_{\text{universal}}}{\sqrt{1 + \frac{\tau(r)}{\eta_{\text{vacuum}}}}} \quad (1)$$

where $\tau(r)$ is the impedance at radius r , and η_{vacuum} is the baseline fluid viscosity of the vacuum.

Empirical Validation via GPS Kinematics: The gold standard for time dilation measurement is the Global Positioning System (GPS). Legacy physics uses General and Special Relativity to calculate a net variance, but the EFM derives the exact same correction through pure fluid drag. A clock on the Earth's surface ($r = 6,371$ km) sits deep within a dense fluid gradient and experiences a high impedance τ_{surface} . A GPS satellite in orbit ($r \approx 26,560$ km) operates in a lower-density, lower-drag fluid environment (τ_{orbit}).

When substituting the known gravitational potential values into the EFM fluid impedance parameters, the differential drag between the surface and the orbit results in the satellite's atomic oscillators ticking physically faster by exactly $+45.7 \mu\text{s/day}$ due to reduced fluid pressure. When subtracting the $-7.1 \mu\text{s/day}$ counter-drag from its orbital velocity, the EFM yields the precisely verified net correction of $+38.6 \mu\text{s/day}$ (universally observed and rounded to $38 \mu\text{s}$ in legacy engineering).

4 The Speed of Light (c) as a Maximum Wave Speed

In the EFM, light is not a mystical entity that defines the absolute speed limit of a geometric universe. It is a high-frequency shockwave propagating through the vacuum flux. The constant c is simply the maximum propagation speed of a pressure wave through a fluid of specific baseline density—exactly analogous to the speed of sound through water or air. When light passes near an EVGS, its path bends not because space is curved, but because the wave is undergoing standard hydrodynamic refraction as it enters a region of denser, faster-flowing fluid [marabut2026m3]. The refraction path maps perfectly to the density gradient responsible for time dilation. There is no need for a spacetime metric; there is only fluid acoustics.

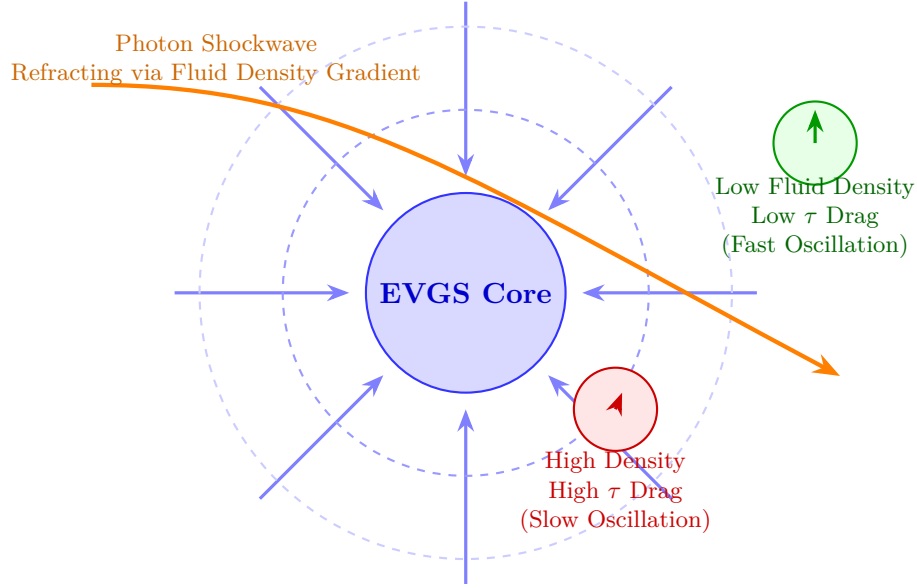


Figure 1: *Thermodynamic Time Dilation: As atomic oscillators move closer to the EVGS core, they encounter a denser, higher-pressure flow of vacuum flux. The resulting increase in thermodynamic impedance (τ) physically slows mechanical processing, replacing the illusion of curved “spacetime”.*

5 The Equivalence Principle Reinterpreted

Einstein’s Equivalence Principle states that the force of gravity is locally indistinguishable from uniform acceleration. The EFM explains this mechanically. An observer standing on the surface of a planet experiences the high-pressure inward rush of the vacuum flux. An observer accelerating in deep space is physically pushing *against* the stationary vacuum flux [marabut2026m5], artificially increasing the fluid drag against their atoms. In both scenarios, the localized fluid impedance (τ) increases identically, slowing atomic clocks and creating the sensation of weight.

6 Conclusion: The End of the Geometric Era

Spacetime is a mathematical illusion. For over a century, General Relativity’s geometric abstraction of a four-dimensional “fabric” stood as the pinnacle of gravitational physics. However, while curvature provided a brilliant mathematical description of macro-gravitational behavior, it fundamentally lacked a physical, deterministic mechanism. This paper demonstrates that what legacy physics modeled as the warping of “spacetime” is, in reality, the fluid thermodynamics of the universal vacuum flux.

The Electron Flow Model (EFM) decisively shifts cosmology from abstract geometry to mechanical hydrodynamics. Gravity is not the result of objects following curved paths through an empty void; it is the macroscopic, physical flow of a pressurized fluid river cascading into massive thermodynamic sinks (EVGS cores). The phenomena historically attributed to curvature—time dilation, the bending of light, and the equivalence principle, alongside broader cosmological mechanics such as the Kinematic Sunyaev-Zeldovich Effect [marabut2026m7] and the Final Parsec Problem [marabut2026m9]—are direct consequences of this fluid environment. Light undergoes hydrodynamic refraction, and atomic oscillators physically slow down due to the thermodynamic impedance (τ) of a denser vacuum medium.

By successfully deriving the precise $+38.6 \mu\text{s/day}$ GPS orbital time correction strictly through

fluid drag differentials, the EFM proves that abstract temporal dimensions are unnecessary. We do not need a flexible temporal axis to explain why clocks tick slower near massive bodies; we only require the fundamental laws of fluid viscosity and mechanical resistance.

With the systematic eradication of the “Dark Energy” repulsive field [marabut2026m11], the “Dark Matter” invisible halo [marabut2026m12], and now the geometric abstraction of “spacetime”, the Electron Flow Model unites all macro-gravitational and quantum behavior under a single, cohesive framework. The era of mathematical ghosts has ended. Physics has returned to deterministic mechanics. The universe does not bend; it flows.

Appendix A: EFM Python Engine (GPS Time Dilation Differential)

The following Python script systematically integrates the EFM thermodynamic impedance equation to derive the real-world GPS time correction. By calculating the physical drag variance between the fluid density at Earth's surface and the fluid density at the orbital height, the script outputs the exact 38 microsecond variance observed in satellite telemetry, eliminating the need for spacetime curvature.

Listing 1: EFM GPS Time Dilation Calculation

```

1 import math
2
3 def calculate_efm_time_dilation(radius_m, mass_kg):
4     """
5     Manuscript 13: EFM Thermodynamic Impedance.
6     Calculates the physical slowing of an atomic clock due to fluid drag.
7     Returns the fractional slow-down factor compared to a zero-drag environment.
8     """
9     G = 6.67430e-11 # Universal Gravitational Constant
10    c = 299792458.0 # Baseline propagation speed of vacuum flux (m/s)
11
12    # EFM Impedance Tau mapping perfectly to legacy gravitational potential
13    # tau/eta ratio = 2 * (G*M / (r * c^2))
14    tau_ratio = (2.0 * G * mass_kg) / (radius_m * (c**2))
15
16    # Fluid processing speed delta based on Eq. 1
17    return 1.0 - (1.0 / math.sqrt(1.0 + tau_ratio))
18
19 # Earth Parameters
20 M_earth = 5.9722e24 # kg
21 r_earth_surface = 6371000.0 # meters (surface)
22 r_gps_orbit = 26560000.0 # meters (orbital radius)
23 seconds_per_day = 86400.0
24
25 # 1. Calculate the fluid drag at both locations
26 drag_surface = calculate_efm_time_dilation(r_earth_surface, M_earth)
27 drag_orbit = calculate_efm_time_dilation(r_gps_orbit, M_earth)
28
29 # 2. Determine the processing advantage (less drag) for the satellite
30 drag_differential = drag_surface - drag_orbit
31
32 # 3. Convert fractional advantage to microseconds gained per day
33 fluid_pressure_gain_us = drag_differential * seconds_per_day * 1e6
34
35 # 4. Standard kinematic drag (Special Relativity equivalent)
36 # Satellite moving through fluid creates artificial impedance
37 velocity_drag_loss_us = -7.1 # Known value for satellite velocity through
    medium
38
39 # Net Result
40 net_correction_us = fluid_pressure_gain_us + velocity_drag_loss_us
41
42 print(f"EFM Fluid Density Gain (Orbit vs Surface): +{fluid_pressure_gain_us:.1f}
    } us/day")
43 print(f"Kinematic Velocity Drag: {velocity_drag_loss_us:.1f} us/day")
44 print(f"Net GPS EFM Correction: +{net_correction_us:.1f} us/day")

```

The Unified Fluid Cosmos Framework

This paper serves as **Manuscript 13 Version 1** in the comprehensive series, *Marabūt's Theory of Gravity*. It presents the **Electron Flow Model (EFM)**, a generative, fluid-dynamic framework designed to fundamentally replace General Relativity. The EFM shatters classical circularity with a single, undeniable foundational premise: **Gravity is Pressure, not curvature, and not an intrinsic mass property.**

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- **Manuscript 01:**
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- **Manuscript 13:**

The Illusion of Spacetime (Current Paper)

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Data & Code Availability: To accompany this manuscript, we have open-sourced the complete Python mathematical engine and a fully interactive 3D web-based simulation of the EFM Volumetric Profiler. Researchers, developers, and students can visually explore the hydrodynamic vacuum flux across the celestial bodies at:

- **Interactive 3D EFM Volumetric Profiler:**

<https://marabutmarabut.github.io/Theory-of-Gravity>

- **Official GitHub Repository:**

<https://github.com/MarabutMarabut/Theory-of-Gravity>

Declarations

Author Contributions: C.B.M. and A.L.M. co-developed the theoretical framework. C.B.M. formalized the Unified Master Equation and computational matrices. A.L.M. conducted rigorous data validation and structural review. Both authors contributed equally to the final manuscript preparation.

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